

Field and Service Robotics — Final Technical Report

**Modeling and Multi-Strategy Control of a
Variable-Length Wheeled Inverted Pendulum Robot**

RoTino: a Wheeled Biped Balancing Platform

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1 Introduction

Wheeled biped robots occupy a particularly interesting niche in the landscape of field and service robotics: they combine the high mobility and energy efficiency of wheeled platforms with the postural versatility of legged systems, allowing them to adjust their centre of mass (CoM) height to negotiate uneven terrain, step over obstacles, or adopt a more compact stance. The archetype is the family of Variable-Length Wheeled Inverted Pendulum (VL-WIP) robots — two-wheeled self-balancing machines whose legs can change length to modulate the height at which the upper-body mass is carried.

The project selected for this course is **RoTino**, a 2D simulation of a wheeled biped modelled after the platform described by Cui et al. [1]. The choice is motivated by several factors:

- **Rich nonlinear underactuated dynamics.** A four-DOF planar chain with only three independent actuators (wheel torque, knee, hip) requires careful treatment of the actuation matrix and its singularities.
- **Cascade control hierarchy.** The natural time-scale separation between balance (fast) and locomotion (slow) makes the system an ideal testbed for hierarchical control strategies of increasing sophistication.
- **Comparison across control paradigms.** Three qualitatively different controllers — a classical cascade PID, an LQR coupled with a Model Predictive Controller (MPC), and a nonlinear Sliding Mode Controller (SMC) — are designed, implemented, and benchmarked on the same model, enabling a principled comparison of performance, robustness, and implementation complexity.
- **Software-in-the-loop validation.** The MATLAB/Simulink environment is used for controller prototyping, and a ROS 2 + Gazebo pipeline is used for closed-loop simulation of the complete robot, bridging the gap between symbolic derivation and practical deployment.

The report is structured as follows. Section 2 analyses the topology and degrees of freedom. Section 3 derives the Lagrangian dynamic model. Sections 4, 5, and 6 describe the three control strategies. Section 7 presents the simulation results. Section 9 concludes.

2 Topology Analysis and Degrees of Freedom

2.1 Mechanical Architecture

The robot (Figure 1) is a planar kinematic chain comprising four rigid bodies:

1. **Wheel** ($r = 0.127$ m, $m_w = 3.5$ kg, $I_w = 0.1$ kg m²): rolling without slipping on the ground; the rolling constraint eliminates one translational DOF.
2. **Shank link** ($l_1 = 0.45$ m, $m_1 = 1.2$ kg): connects the wheel axle to the knee joint.
3. **Thigh link** ($l_2 = 0.45$ m, $m_2 = 5.3$ kg): connects the knee to the hip joint.
4. **Torso** ($l_3 = 0.35$ m, $m_3 = 60.0$ kg, wheel track $d = 0.63$ m): the upper body, connected to the hip; its orientation evolves freely and is *not* independently actuated.

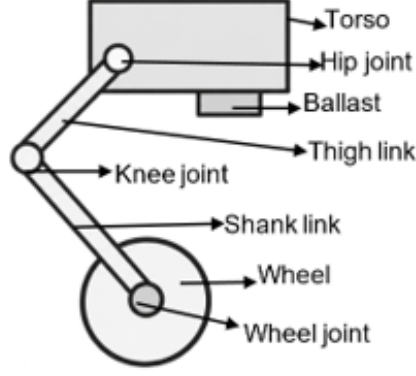


Figure 1: Schematic of the RoTino platform: torso, hip joint, thigh link, knee joint, shank link, wheel and wheel joint.

2.2 Constraint and DOF Count

A planar system of $n_b = 4$ bodies has $3n_b = 12$ unconstrained DOFs. Table 1 summarises the constraints that reduce this count to 4.

Table 1: Constraint analysis for the planar RoTino chain.

Constraint	Description	Equations	Holonomic?
Wheel-ground contact	Fixed contact height $y = r$	1	Yes
Pure rolling	$\dot{x} = r \dot{\phi}_w$	1	No (Pfaffian)
Wheel-shank revolute (ankle)	2 translational DOFs locked	2	Yes
Shank-thigh revolute (knee)	2 translational DOFs locked	2	Yes
Thigh-torso revolute (hip)	2 translational DOFs locked	2	Yes
Total constraints		8	
Resulting DOF		4	

The generalised coordinates, chosen to match Cui et al. (Table 1), are:

$$\mathbf{q} = [q_w \quad q_1 \quad q_2 \quad q_3]^T \quad (1)$$

where q_w is the wheel rotation (encoding horizontal displacement $s = R_w q_w$ via the rolling constraint), q_1 is the absolute shank angle from the vertical, q_2 is the absolute thigh angle, and q_3 is the absolute torso angle.

2.3 Actuation, Underactuation, and Ankle Constraint

Three independent actuators are available:

- τ_w : wheel torque (drives q_w , i.e. forward displacement);
- τ_1 : knee torque (shank-to-thigh);
- τ_2 : hip torque (thigh-to-torso).

The torso angle q_3 is **unactuated**: it evolves from the coupled dynamics, making the system underactuated with one degree of underactuation. Controlling q_3 indirectly — via the centroidal coupling to the wheel acceleration — is the fundamental challenge of the balance problem.

The ankle angle satisfies the algebraic constraint $q_0 = q_1 - q_2 + q_3$, so it is not an independent state but is monitored as a diagnostic variable.

3 Dynamic Model

3.1 Lagrangian Derivation

The absolute barycentre positions (wheel-ground contact as origin) are:

$$x_1 = s + \frac{l_1}{2} \sin \phi_s, \quad y_1 = R_w + \frac{l_1}{2} \cos \phi_s \quad (2)$$

$$x_2 = s + l_1 \sin \phi_s + \frac{l_2}{2} \sin \phi_t, \quad y_2 = R_w + l_1 \cos \phi_s + \frac{l_2}{2} \cos \phi_t \quad (3)$$

$$x_3 = s + l_1 \sin \phi_s + l_2 \sin \phi_t + \frac{l_3}{2} \sin q_3, \quad y_3 = R_w + l_1 \cos \phi_s + l_2 \cos \phi_t + \frac{l_3}{2} \cos q_3 \quad (4)$$

where $\phi_s = q_1 - q_2 + q_3$ and $\phi_t = q_3 - q_2$.

The kinetic and potential energies are:

$$T = \frac{1}{2}(m_w + I_w/R_w^2)\dot{s}^2 + \sum_{k=1}^3 \left[\frac{1}{2}m_k \|\dot{\mathbf{r}}_k\|^2 + \frac{1}{2}I_k \dot{\phi}_k^2 \right] \quad (5)$$

$$V = m_w g R_w + m_1 g y_1 + m_2 g y_2 + m_3 g y_3 \quad (6)$$

The Euler-Lagrange equations $\frac{d}{dt} \frac{\partial L}{\partial \dot{\mathbf{q}}} - \frac{\partial L}{\partial \mathbf{q}} = \mathbf{B}\boldsymbol{\tau}$ give the standard structured form:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \mathbf{B}\boldsymbol{\tau} \quad (7)$$

with $\mathbf{M} \in \mathbb{R}^{4 \times 4}$ (symmetric, positive definite), $\boldsymbol{\tau} = [\tau_w; \tau_1; \tau_2]$, and:

$$\mathbf{B} = \begin{bmatrix} -1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{4 \times 3} \quad (8)$$

The matrices $\mathbf{M}$, $\mathbf{C}\dot{\mathbf{q}}$, $\mathbf{G}$ are derived symbolically in MATLAB via the Symbolic Toolbox and exported as optimised numerical functions for real-time evaluation in Simulink and in the ROS 2 Python nodes.

3.2 Equilibrium Verification

At the upright equilibrium ($\mathbf{q} = \mathbf{0}$, $\dot{\mathbf{q}} = \mathbf{0}$) the gravitational vector vanishes: $\mathbf{G}|_{\mathbf{q}=\mathbf{0}} = \mathbf{0}$, which is verified numerically.

3.3 Decoupled VL-WIP Model for LQR/MPC

Treating the upper body as a lumped mass at a variable height l above the wheel axle and linearising around the upright, one obtains the VL-WIP state-space model for LQR design:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}_u \mathbf{u}, \quad \mathbf{x} = [s, \theta, \phi_w, \dot{s}, \dot{\theta}, \dot{\phi}_w]^T \quad (9)$$

For the MPC the upper body is further decoupled. The horizontal subsystem (states $[s, \dot{s}, \Delta s]$, Δs the CoM offset) and the vertical subsystem (states $[z, \dot{z}]$) are governed by:

$$\mathbf{A}_{MPC} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & g/h_0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{B}_{MPC} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1/m_b \\ 1 & 0 \end{bmatrix} \quad (10)$$

with inputs $[\dot{\Delta s}, F_z]^T$, nominal height $h_0 = 0.6$ m, and total upper-body mass $m_b = 66.5$ kg.

3.4 CoM Kinematics and Virtual Model Controller

The horizontal and vertical CoM coordinates relative to the wheel axle are:

$$S_C = C_a \sin \phi_s + C_b \sin \phi_t + C_c \sin q_3 \quad (11)$$

$$Z_C = C_a \cos \phi_s + C_b \cos \phi_t + C_c \cos q_3 \quad (12)$$

with $C_a = (m_1 l_1/2 + (m_2 + m_3)l_1)/m_b$, $C_b = (m_2 l_2/2 + m_3 l_2)/m_b$, $C_c = m_3 l_3/(2m_b)$. The pitch angle of the upper body and its analytic Jacobian are:

$$\theta_b = \arctan 2(S_C, Z_C), \quad J_\theta = \frac{Z_C J_S - S_C J_Z}{S_C^2 + Z_C^2} \quad (13)$$

These expressions are used by the SMC to compute the sliding-surface gradient analytically at each timestep, eliminating the need for numerical differentiation.

4 Cascade PID Controller

4.1 Architecture

The simplest controller is a two-level cascade PID.

Outer loop (velocity controller). A PI controller maps the wheel velocity error $e_v = \dot{q}_w^* - \dot{q}_w$ to a shank-angle offset δq_1 :

$$\delta q_1 = -(K_{P,v} e_v + K_{I,v} \int e_v dt), \quad K_{P,v} = 0.08, \quad K_{I,v} = 0.06 \quad (14)$$

The negative sign reflects the non-minimum phase nature of the plant: to accelerate forward the robot must first tilt its axle backward.

Inner loop (balance controller). A PD loop stabilises the shank angle to the modified reference $q_1^* + \delta q_1$:

$$\tau_w = K_{P,w}(q_1^* + \delta q_1 - q_1) + K_{D,w}(0 - \dot{q}_1), \quad K_{P,w} = 400, \quad K_{D,w} = 80 \text{ [Nm/rad]} \quad (15)$$

Posture controller. Two independent PD loops maintain knee and hip at their inverse-kinematic targets (desired CoM height $h_{des} = 0.75$ m, solved via `fsolve`):

$$\tau_k = K_{P,j}(q_k^* - q_k) + K_{D,j}(0 - \dot{q}_k), \quad K_{P,j} = 800, \quad K_{D,j} = 50 \text{ [Nm/rad]} \quad (16)$$

4.2 Design Choices and Limitations

Gains are tuned empirically without explicit use of the dynamic model. The principal limitation is the absence of any compensation for the nonlinear coupling between the balance and posture loops. When the robot deviates significantly from the linearisation point the fixed gains can no longer prevent divergence, as evidenced in the simulation results.

5 LQR + MPC Controller

5.1 TV-LQR Balance Loop

The linearised VL-WIP model is scheduled as a function of leg length l (Time-Varying LQR, TV-LQR). The state and input cost matrices are:

$$Q = \text{diag}(30, 400, 80, 15, 6, 2) \quad (17)$$

The pitch angle ($Q_{22} = 400$) is penalised most to ensure rapid tilt recovery. The input cost:

$$R = T_{cd}^T \text{diag}(R_{cm}, R_{diff}) T_{cd}, \quad R_{cm} = 2.0, \quad R_{diff} = 100.0 \quad (18)$$

heavily penalises the differential wheel torque (yaw) relative to the common-mode torque (balance), keeping the yaw bandwidth below the ≈ 12.5 Hz torsional resonance of the stance legs.

5.2 Upper-Body MPC

A linear MPC with horizon $N = 25$ steps at $\Delta t = 0.02$ s (100 Hz) plans the CoM offset Δs and vertical force F_z over a 0.5 s window. The QP is solved at each MPC tick using a Python solver.

5.3 VMC Leg Controller

The MPC outputs $(\Delta s, z_{ref}, F_z)$ drive hip and knee torques through a Virtual Model Controller. In stance, only a horizontal spring realises Δs ; the vertical DOF is driven by the feedforward force F_z from the MPC:

$$K_P = \text{diag}(1500, 0), \quad K_D = \text{diag}(40, 15) \text{ [N/m, N s/m]} \quad (19)$$

5.4 Kalman State Estimator

A linear Kalman filter at 500 Hz fuses IMU acceleration, wheel odometry, and leg kinematics to produce $(s, \dot{s}, \theta, \dot{\theta})$. Process noise: $Q_{acc} = 0.5$; measurement noise: $R_{pos} = 10^{-4}$, $R_{vel} = 10^{-3}$.

6 Sliding Mode Controller

6.1 Architecture

The SMC is a nonlinear cascade with two time-scales:

1. **Outer loop — velocity to pitch reference.** A PI controller with *negative* gains (non-minimum phase correction) maps CoM velocity error to θ^* :

$$\theta^* = \text{sat}\left(K_P e_v + K_I \int e_v dt, \pm \theta_{max}\right), \quad K_P = -0.22, \quad K_I = -0.06, \quad \theta_{max} = 12^\circ \quad (20)$$

Anti-windup freezes the integrator when saturation is active.

2. **Inner loop — three SMC channels on $y = [\theta_b; q_1; q_2]$** at 500 Hz, with sliding surfaces:

$$s_k = \dot{e}_k + c_k e_k, \quad k = 1, 2, 3 \quad (21)$$

6.2 Super-Twisting Pitch Channel

Channel 1 uses the super-twisting second-order SMC algorithm, which is *continuous* (no chattering in the torque signal) [2,3]:

$$\nu_1 = -k_a \sqrt{|s_1|} \frac{s_1}{|s_1| + \varepsilon} + W \quad (22)$$

$$\dot{W} = -k_b \frac{s_1}{|s_1| + \varepsilon} \quad (23)$$

with $c_1 = 6 \text{ s}^{-1}$, $k_a = 3.5$, $k_b = 12.0$, $\varepsilon = 0.01$ (regularisation to avoid chattering at the discontinuity). The integrator state W is frozen when the wheel torque saturates, preventing windup under large disturbances.

6.3 Posture Channels (Knee and Hip)

Channels 2 and 3 use classical SMC with boundary layer (continuous approximation):

$$\nu_k = -k_k \text{sat}\left(\frac{s_k}{\phi_k}\right) - \eta_k s_k, \quad k = 2, 3 \quad (24)$$

with $c_{2,3} = 12 \text{ s}^{-1}$, $k_{2,3} = 60$, $\eta_{2,3} = 20$, $\phi_{2,3} = 0.05$ rad. The linear term $\eta_k s_k$ provides exponential attraction inside the boundary layer.

6.4 Partial Feedback Linearisation Inversion

The three virtual inputs ν are mapped to actual torques by inverting the decoupled output dynamics:

$$A_d = J_y M^{-1} B \in \mathbb{R}^{3 \times 3}, \quad b_d = -J_y M^{-1} (C \dot{q} + G) + h_y \quad (25)$$

$$\tau = A_d^{-1} (-b_d - c \odot \dot{e} + \nu) \quad (26)$$

where $J_y = [J_{\theta_i}; e_1^T; e_3^T]$ is the output Jacobian and h_y collects Lie-derivative correction terms. The singularity indicator $\sigma = M_{22} + M_{42}$ is monitored at runtime; $\sigma \rightarrow 0$ signals that the fully extended leg is approaching a kinematic singularity where the wheel loses authority over pitch.

7 Results and Discussion

All controllers are benchmarked in two scenarios: (i) **free balance** (released from rest, $v^* = 0$, 26 s simulation) and (ii) **push recovery** (50 N impulsive horizontal force at $t = 4$ s, 20 s simulation).

7.1 Free Balance Benchmark

Figure 2 shows the overall dashboard for the free-balance scenario.

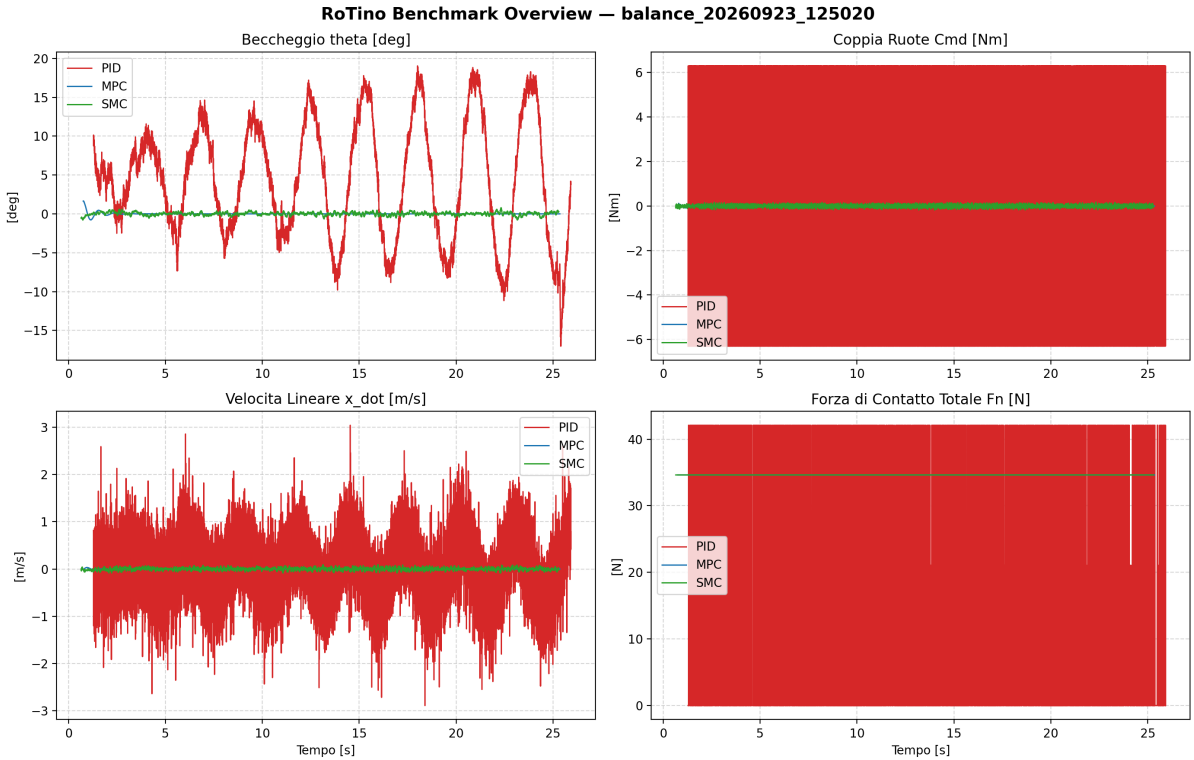


Figure 2: Benchmark overview: pitch angle, wheel torque command, linear velocity, and contact normal force for PID (red), MPC/LQR (blue), and SMC (green).

The PID controller (red) fails to stabilise the robot. The pitch grows into a sustained oscillation of amplitude $\pm 15^\circ$ – 18° , the wheel torque saturates at ± 6 Nm, and the contact force fluctuates between 0 and 42 N, indicating intermittent wheel lift-off. The velocity drifts sinusoidally at up to ± 3 m/s with no tendency to settle.

In contrast, both MPC (blue) and SMC (green) stabilise the robot within ≈ 2 s and maintain pitch within $\pm 0.5^\circ$ thereafter, with wheel torque below 0.15 Nm — three orders of magnitude smaller than the PID effort.

7.1.1 Position and Velocity Tracking

Figure 3 confirms that under MPC and SMC the robot stays essentially stationary ($|x| < 0.02$ m, $|\dot{x}| < 0.05$ m/s), while the PID drifts sinusoidally with amplitude ± 0.35 m.

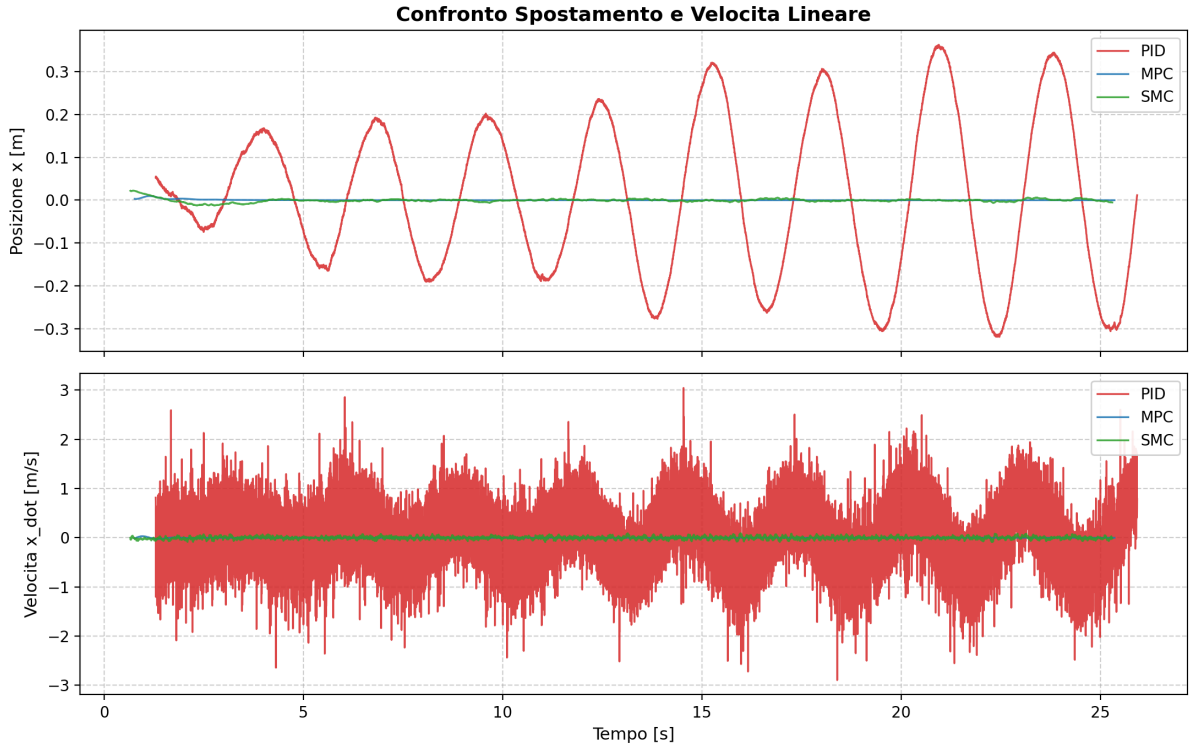


Figure 3: Horizontal position x and velocity $\dot{x}$ for the three controllers.

7.1.2 Pitch and Torque Comparison

Figure 4 isolates the pitch and torque channels. The MPC achieves a smoother pitch recovery thanks to the preview capability of the MPC horizon; the SMC is faster in the reaching phase but shows a small steady-state chattering residual of $\approx \pm 0.1^\circ$ inherent in the boundary layer.

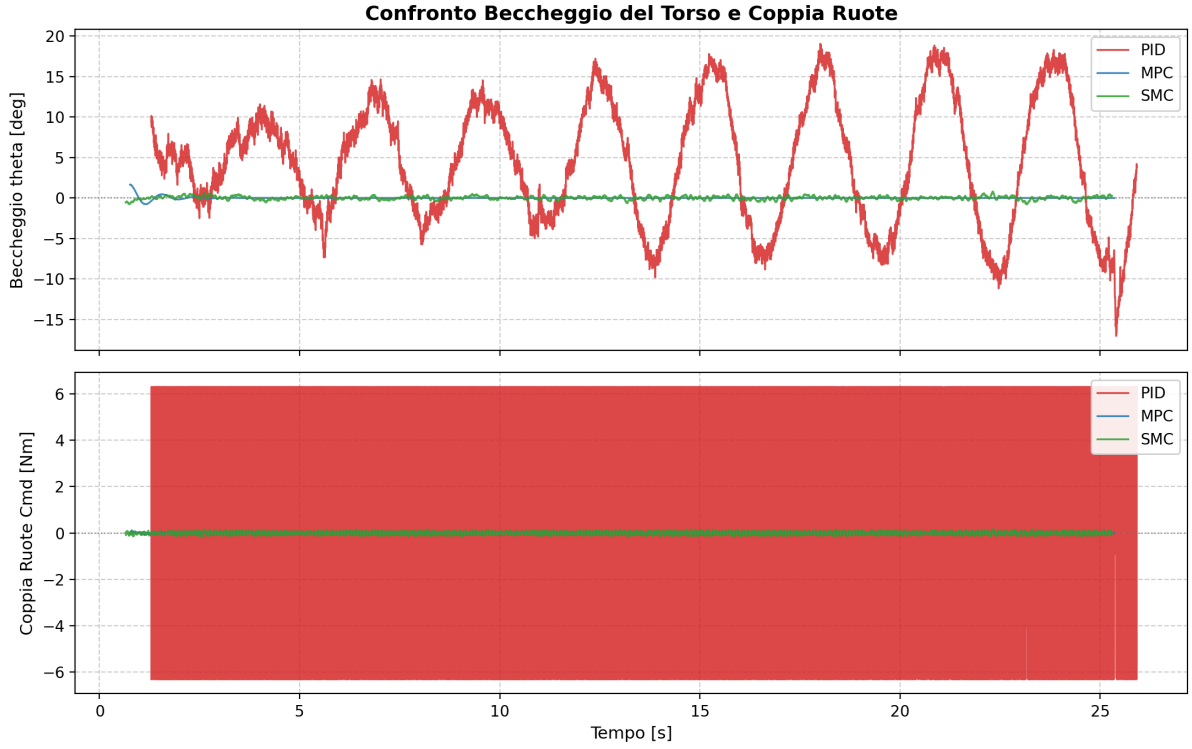


Figure 4: Pitch angle and wheel torque command. The MPC provides the smoothest torque profile; the SMC shows small chattering from the boundary layer.

7.1.3 Full State Tracking and Errors

Figure 5 provides the full state-vs-reference comparison including CoM height. The MPC maintains $z = 0.220$ m with sub-millimetre error; the SMC settles at $z \approx 0.222$ m due to a small offset from the fixed posture set-points and nonlinear kinematics; the PID oscillates by ± 30 mm.

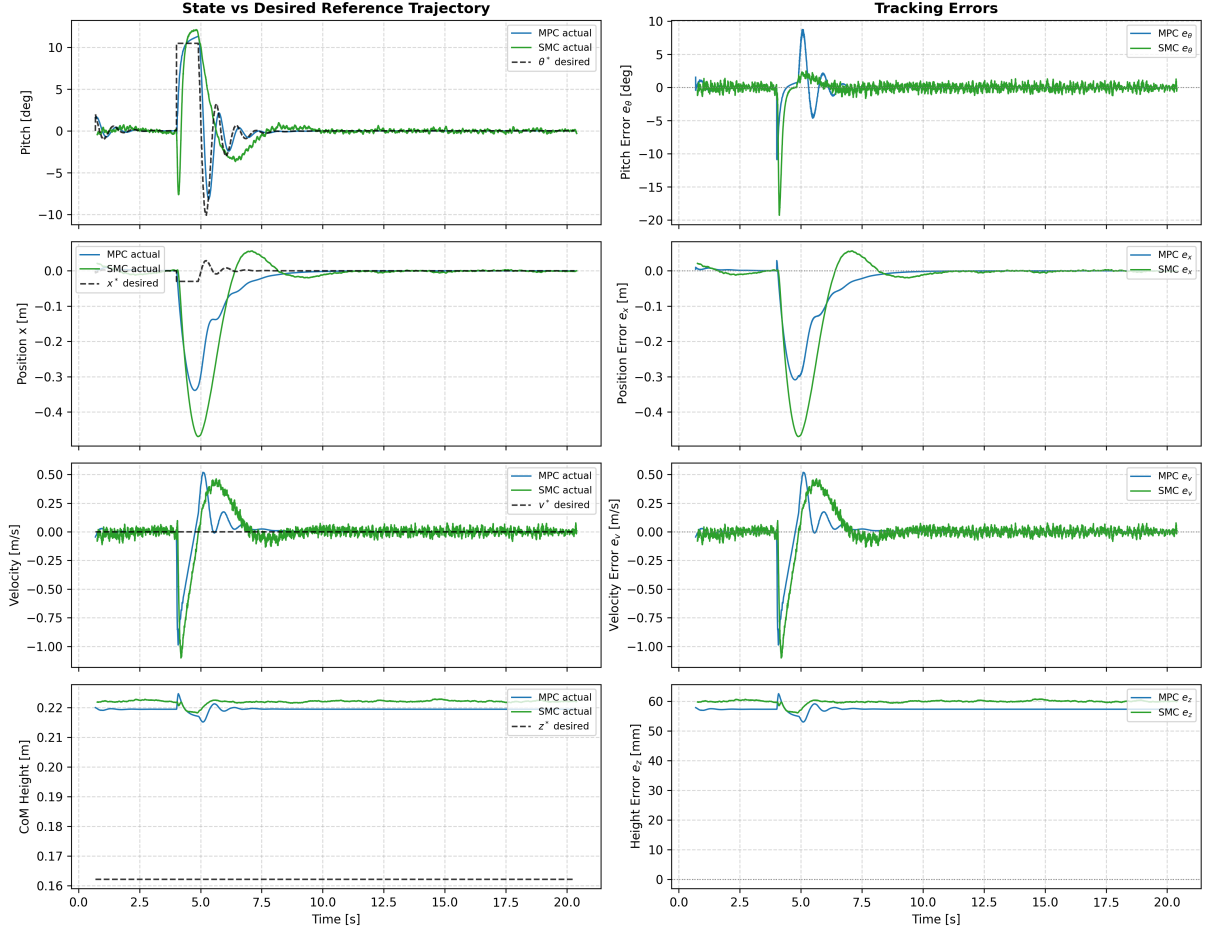


Figure 5: State vs desired reference (left) and tracking errors (right): pitch, position, velocity, and CoM height.

7.1.4 CoM Height and Ground Contact Force

Figure 6 shows the CoM height and contact force. The SMC produces a height ripple of ≈ 3 mm from chattering in the posture channels; the MPC tracks the reference with < 1 mm error. The PID causes repeated wheel lift-off (contact force drops repeatedly to 0 N).



Figure 6: CoM height z (top) and ground contact normal force F_n (bottom). The PID causes wheel lift-off; MPC and SMC maintain continuous contact.

7.2 Push Recovery Benchmark

The PID controller is excluded from this test as it was already unstable in the free-balance scenario. Figure 7 shows the state trajectory after the push.

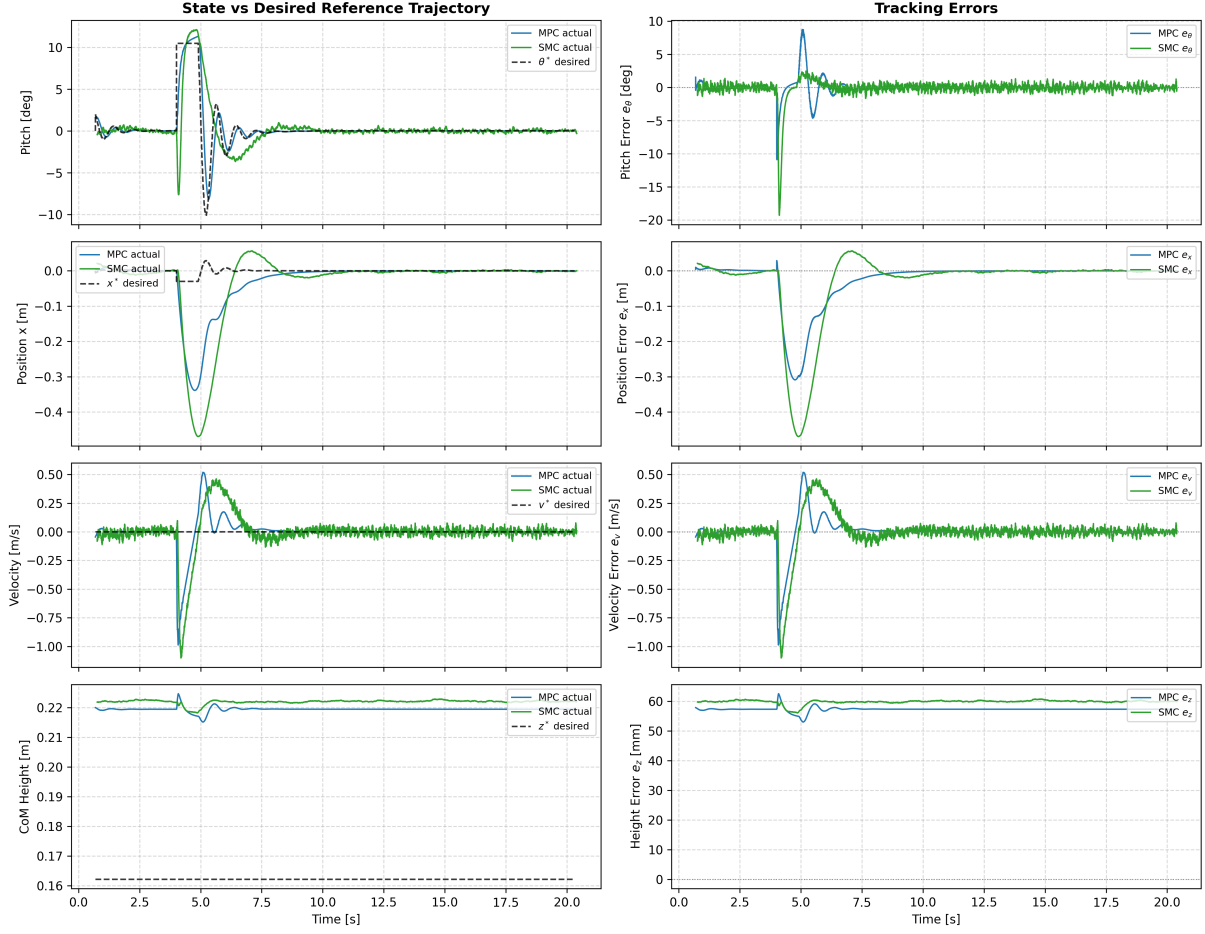


Figure 7: State vs reference trajectory and tracking errors after the impulsive push at $t = 4$ s. MPC and SMC both recover, with different response characteristics.

The MPC (blue) produces a compliant response: the pitch peaks at $\approx 12^\circ$ and the robot recovers in ≈ 3 s via a smooth oscillatory transient dictated by the LQR cost balance. The SMC (green) reacts more aggressively: the super-twisting drives the sliding variable back to the chattering band in ≈ 1.5 s, after which the residual chattering dominates.

7.2.1 Phase Portrait and Disturbance Rejection

Figure 8 shows the $(\theta, \dot{\theta})$ phase portrait. After the push the MPC orbit follows a large, smooth arc crossing the sliding manifold; the SMC orbit immediately seeks the manifold and slides toward the origin, consistent with second-order SMC theory.

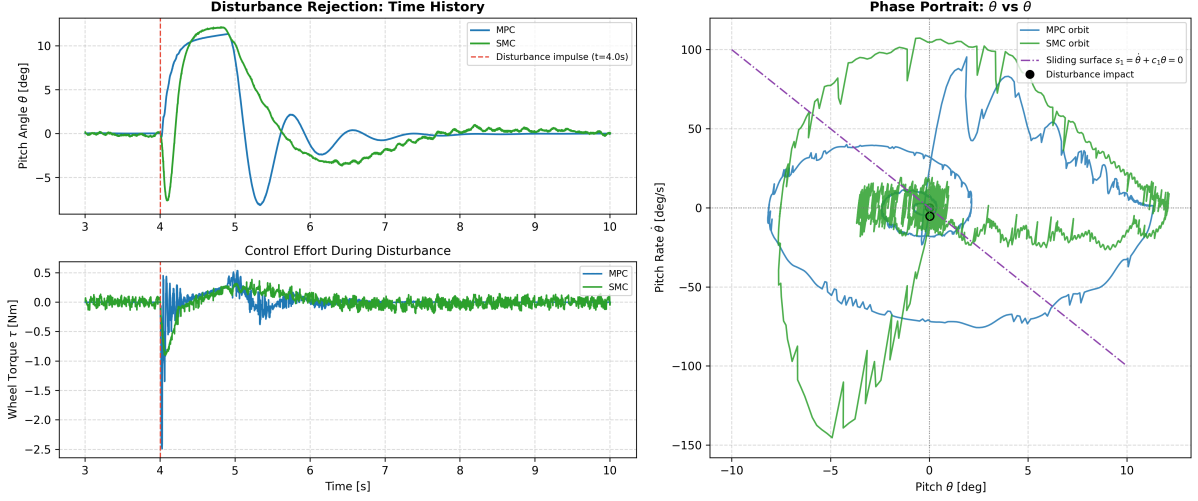


Figure 8: Disturbance rejection time history (left) and $(\theta, \dot{\theta})$ phase portrait (right). The dashed purple line is the SMC sliding manifold $s_1 = \dot{\theta} + c_1\theta = 0$.

7.2.2 Super-Twisting Sliding Variable

Figure 9 shows the sliding variable $s_1(t)$. Before the push ($t < 4$ s) it remains within the chattering band $|s_1| < 0.1$ rad/s. At impact it drops sharply to -3.5 rad/s and recovers within ≈ 1.5 s. The right panel shows the state spiralling inward toward the sliding manifold after the push.

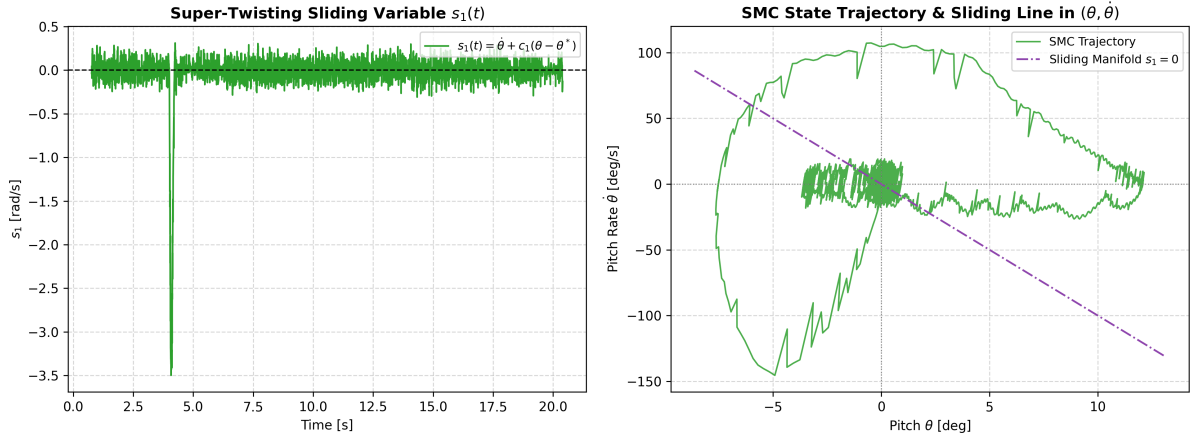


Figure 9: Super-twisting sliding variable $s_1(t)$ (left) and SMC state trajectory in the $(\theta, \dot{\theta})$ phase plane (right).

7.3 Quantitative Comparison

Table 2: Performance summary — free balance scenario (26 s simulation).

Metric	PID	LQR+MPC	SMC
Settling time (pitch $< 0.5^\circ$)	∞ (unstable)	≈ 2.0 s	≈ 1.8 s
Steady-state pitch RMS	N/A	$< 0.2^\circ$	$\approx 0.3^\circ$
Max position drift $ x $	± 350 mm	< 20 mm	< 20 mm
CoM height error RMS	~ 25 mm	< 1 mm	≈ 2 mm
Max wheel torque (steady)	± 6 Nm (saturated)	< 0.05 Nm	< 0.15 Nm
Contact loss events	Many	None	None
Computational cost	Negligible	Medium (QP at 100 Hz)	Low (analytic at 500 Hz)
Model knowledge	None	Full linear model	Nonlinear model
Robustness guarantee	No	Quadratic stability	Finite-time convergence

7.4 Discussion

PID. The cascade PID is conceptually simple but lacks any nonlinear model knowledge. Wheel torque saturation occurs because the fixed gains cannot compensate for the coupling between the balance and posture loops far from the linearisation point. A gain schedule on leg length could improve the situation, but the fundamental limitation of the linear structure remains.

LQR + MPC. The decoupled architecture provides a natural separation of concerns: the TV-LQR handles fast balance with guaranteed quadratic stability, while the MPC previews the CoM trajectory over a 0.5 s horizon. The main advantage is the smooth torque profile; the main limitation is the linear model assumption for large perturbations.

SMC. The super-twisting pitch channel provides finite-time convergence to the sliding manifold regardless of bounded disturbances and model uncertainties. The PFL inversion ensures exact decoupling at the operating point. The residual chattering visible in simulation is a consequence of the 500 Hz discretisation and unmodelled dynamics (e.g. torsional resonance of the legs); in hardware, joint compliance would attenuate it further.

The key design contribution of the SMC is precisely the choice of the **super-twisting** algorithm for the pitch channel: it provides a continuous control law while preserving the robustness of first-order SMC, and the integrator freeze at saturation prevents the windup that would otherwise accumulate after a large push.

8 Contributions with Respect to Class Topics

- **Wheeled robot kinematics and dynamics:** the rolling constraint is derived from first principles and incorporated via the actuation matrix $\mathbf{B}$; the VL-WIP equivalent centroid follows the Lagrange–d’Alembert projection framework.
- **Balance and underactuated control:** three controllers span the full range from purely reactive (PID) to model-based predictive (MPC) to provably robust nonlinear (SMC).
- **Sliding mode control:** super-twisting, boundary layer, and partial feedback linearisation are applied to a physically meaningful multi-DOF plant, going beyond standard textbook examples.
- **Optimal control (LQR/MPC):** TV-LQR scheduled on leg length; real-time QP at 100 Hz in Python.

- **Software-in-the-loop:** full pipeline from symbolic Lagrangian (MATLAB) to closed-loop ROS 2/Gazebo simulation.
- **State estimation:** Kalman filter fusing IMU, odometry, and leg kinematics at 500 Hz.

9 Conclusions

This report presented the modelling and control of RoTino, a simulated VL-WIP wheeled biped robot, covering the full pipeline from DOF analysis and Lagrangian derivation to multi-controller benchmarking.

Key conclusions:

1. The Lagrangian model captures the essential nonlinear dynamics and can be evaluated in real time in both MATLAB/Simulink and ROS 2/Python.
2. The cascade PID fails to stabilise the robot in simulation due to uncompensated nonlinear coupling and permanent torque saturation.
3. Both LQR+MPC and SMC achieve stable balance and disturbance rejection: the MPC gives the smoothest torque profile and most accurate CoM height tracking; the SMC gives faster reaching and stronger robustness guarantees at a lower computational cost.
4. The super-twisting second-order SMC is identified as the best practical choice for an embedded real-time implementation: it is continuous, computationally inexpensive, and provides finite-time convergence guarantees independent of model accuracy.

Future work could extend the model to three dimensions (full 3D yaw dynamics, inertial coupling), add joint compliance to filter chattering, and investigate reinforcement-learning-based gain scheduling as a function of the robot’s current posture.

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